Dr. Snigdha Sen

Senior Assistant Professor

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Scopus Link: https://www.scopus.com/authid/detail.uri?authorId=57223155667

Google Scholar: https://scholar.google.com/citations?user=u5dq7NMAAAAJ&hl=en

Enthusiastic Senior Academician having around 15 years of teaching experience with 7 years of dedicated research well-versed in innovative teaching methods and having expertise in conducting workshops in Machine Learning, Deep Learning, Big data and Cloud Computing.

Domains

Computer Vision, Deep Learning, Space data Analysis, Big data analytics and Cloud Computing

Work History

March, 2024 - Present
Senior Assistant Professor
Manipal Institute of Technology Bengaluru, MAHE

August, 2011- February, 2024
Associate Professor
GLOBAL ACADEMY OF TECHNOLOGY, Bangalore, Karnataka

Feb 2010 - June 2010
Programmer Analyst Trainee
Cognizant Technology Solutions, Kolkata, West Bengal

July 2005- July 2007
Lecturer, Computer Science
Aryabhatta Institute of Engineering & Management, Durgapur, West Bengal

Education

2019- 2024
PhD: Data Science & Deep Learning
Indian Institute of Information Technology – Allahabad

2007-2009 **MTech: Information Technology**

Jadavpur University – Kolkata, West Bengal

2001-2005 **B.E: Information Technology**

University Institute of Technology – University of Burdwan, West Bengal

Publication

International Journals

- Dabhade, Aditya, Soumendu Chakraborty, and Snigdha Sen. "A novel block based preparation network using convolutional neural networks for image steganography and robust data protection." *Multimedia Tools and Applications* 85.5 (2026): 482. https://doi.org/10.1007/s11042-026-21652-5 **Scopus Indexed)**
- Sk, A., Chakraborty, S., & **Sen, S. (2025**). A novel blocked based U-Net model for image steganography. *Multimedia Tools and Applications*, 1-20. (**Scopus Indexed)**
- **Sen, Snigdha, et al**. "Performance evaluation of efficient interpretable CNN-transformer model for redshift prediction." *Signal, Image and Video Processing* 19.4 **(2025**): 326. **(SCI Indexed and Scopus Indexed)**
- Mahesh, D., Raju, N.V. & **Sen, S**. Machine learning approaches for predicting dielectric properties of banana fibers reinforced with polypropylene composites. Multiscale and Multidiscip. Model. Exp. and Des. 8, 71 **(2025).** https://doi.org/10.1007/s41939-024-00666-2 (**Scopus Indexed)**
- **Sen, Snigdha,** and Shreya Banerjee. "Precise Image Segmentation using Explainable U-Net based Hybrid Approach for Improved Detection of Skin Lesion." *International Journal of Computing and Digital Systems* 18.1 **(2025).**
- Madhavi, S., Mahesh, D., Raju, N. V., **Sen, S.,** & Johns, J. (2025). Analysing and predicting dielectric properties of bamboo fiber filled polypropylene composites using artificial neural network. Journal of the Indian Academy of Wood Science, 22(1), 37-49. **(Scopus Indexed)**
- **Snigdha Sen**, Krishna Pratap Singh, Pavan Chakraborty, Dealing with imbalanced regression problem for large dataset using scalable Artificial Neural Network, New Astronomy, 2022, (**SCI Indexed and Scopus Indexed)**
- **Sen, S.,** Agarwal, S., Chakraborty, P. et al. Astronomical big data processing using machine learning: A comprehensive review. Exp Astron (2022). (**SCI Indexed and Scopus Indexed)**
- Doddashamachar M, **Sen S,** Nama Vasudeva Setty R. A novel Artificial Neural Network-based model for predicting dielectric properties of banana fiber filled with polypropylene composites. Journal of Thermoplastic Composite Materials. 2023;0(0). doi:10.1177/08927057221148455 (**SCI Indexed and Scopus Indexed)**

International Conferences

- Saha, Jayita, et al. "Extensive Log Analysis Using Small Language Models in Minimal Resource Environments." 2025 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS). Vol. 1. IEEE, 2025.
- Parvez, Sk Sahil, et al. "Efficient LLM Framework Using LORA Based Fine Tuning to Analyze Textual Astronomical Data." International Conference on Soft Computing and its Engineering Applications. Cham: Springer Nature Switzerland, 2025.
- Tufani, R., Sen, S., & Chakraborty, P. (2025, August). Harnessing The Potential of Pandas and PySpark: An In-Depth Performance Evaluation for Machine Learning Classification on Single Node Architectures. In *2025 International Conference on Artificial Intelligence and Machine Vision (AIMV)* (pp. 1-6). IEEE.
- Hassan, A., Juturu, A., Sen, S., & Dsouza, R. (2024, March). Real-Time Object Detection and Tracking Using Mobinet and SSD Algorithms. In *International Conference on Machine Learning, IoT and Big Data* (pp. 449-462). Singapore: Springer Nature Singapore.
- Tufani, R., Sen, S., & Chakraborty, P. (2024, March). Comprehensive Performance Analysis of PySpark and Pandas for Classification and Clustering Task. In *International Conference On Health Informatics, Intelligent Systems And Networking Technologies* (pp. 251-262). Singapore: Springer Nature Singapore.
- J. Saha, M. Kumar, A. Shah and S. Sen, "Extensive Log Analysis Using Small Language Models in Minimal Resource Environments," *2025 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)*, Kuala Lumpur, Malaysia, 2025, pp. 500-505, doi: 10.1109/I2CACIS65476.2025.11100950.
- Snigdha Sen, S Saha, P Chakraborty, KP Singh, "Photometric redshift forecasting using Lipschitz Adaptive Learning Rate", in the 29th International Conference on Neural Information Processing, (ICONIP 2022), IIT-Indore, November 2022 **(CORE-B)**
- Sen, S., Chakraborty, P., Das, S., Pandey, K., & Narayana, P. (2024, July). Investigation of Clustering Methods for SDSS Galaxy Images through Feature Extraction with VGG-16. In 2024 IEEE Space, Aerospace and Defence Conference (SPACE). IEEE.
- Manasa M**, Snigdha Sen "**Interpretable Predictions for crime categories using log loss approach for imbalanced target feature**",** IEEE Conference, InC4, March 2024
- **Sen, Snigdha,** and Pavan Chakraborty. "Evaluating Efficacy of MAML Based Approach on Regression Using Astronomical Imbalanced Dataset." 2024 International Conference on Computational Intelligence and Network Systems (CINS). IEEE, 2024.

Book Chapters

- Hegde, M. I., Dhanvina, N., Dharshan, N., & Sen, S. (2025). Investigation of Machine Learning Algorithms in Detecting Chronic Kidney Disorder. In Machine Learning for Disease Detection, Prediction, and Diagnosis: Challenges and Opportunities (pp. 207-224). Singapore: Springer Nature Singapore.

- Sen, S., & Madhu, B. (2025). Revolutionizing Quantum Cryptography With Artificial Intelligence: A Perspective for Collaborative Research. In *Advancing Cyber Security Through Quantum Cryptography* (pp. 441-464). IGI Global Scientific Publishing.

FDP Attended (2024-2025)

- 5 Days FDP on Next Generation Computing Paradigms by MIT Bengaluru, February 2026
- 5 Days FDP on "Innovations in Space and defense" by RIJIT, November 2025
- 5-day Faculty Development Program on Extended Reality and Quantum Computing Advancing Science, Industry and Game Development (Unity) by MIT Bengaluru, June 2025
- 2 Days Workshop on "Design Thinking" by MIT Bengaluru, March 2025
- 5 Days FDP on AI/ML Applications in Astronomy & Astrophysics (AMAA] by IUCAA Pune January 2025
- 3 Days FDP on "LLM Model" by IISC Bengaluru, March 2025
- 5 Days FDP on "Bridging the Gap between AI and Healthcare" by IISC Bengaluru, May 2025

Awards/Achievements

- Selection of the Project titled **Care-In-A-Cup: A Solution to Tend to the Needs of Differently Abled Women** at state level exhibition by KSCST and VTU
- Winner in Paper presentation event (**Title: Working with Docker: The New Industry Need**) at 34th CSI Karnataka State Student Convention held at KSIT, Bangalore, December 2020
- Winner at National level Ideathon 2021 organized by VVCE IEEE COMPUTER SOCIETY STUDENT CHAPTER (Title: **Preserving the Street Vendor tradition of India and optimizing their profit post Covid19- A Web App using Data Analytics**), June 2021

Grant/Funding/Patents

Title: Constructing small language models as SME using minimal resources
Role: Mentor/Project Guide
Duration: 2023-24
Agency: KSCST
Amount: 4000/

Title: Quantitative analysis of prospective data to predict the likelihood of PCOD based on physical and psychological factors that affect young adult women using Machine Learning

Role: CO-PI
Duration: 2019-20
Agency: VGST
Amount: 3 Lakh

Title: Care-In-A-Cup: A Solution to Tend to the Needs of Differently Abled Women
Role: Mentor/Project Guide
Duration: 2021-22
Agency: KSCST
Amount: 3500/

Proposal Under evaluation

- ✓ Investigation of the Baby Dragon Hatchling (BDH) architecture for satellite remote sensing image applications (Submitted to ISRO)
- ✓ A Data-Driven and Sustainable AI/ML Solution for Enhancing Rural Education in Karnataka (Submitted to VGST)
- ✓ Digital Preservation of Indian Ancient Heritage Using Artificial Intelligence and Immersive technologies: A Metaverse Approach

Patents:

- ✓ **Title of the invention:** a system and method for multimodal on-device emergency detection with indoor localization and alerts **(Published)**
- ✓ **Title of the invention:** a system and method for optimizing emergency medical response through real-time data integration **(Applied)**